Feedback + next step :

I had a look at your two notebooks (`EDA_Assignement_CD4_TYF` and `Simple_ABM_CD4_TYF / ABM.ipynb`). Here is my feedback and the next step.

1. Feedback on your current model (Simple_ABM_CD4_TYF)

Positive points

- You correctly implemented the **4-compartment model** with:
  - `N` (naive), `S` (TSCM), `C` (TCM), `R` (TEMRA),
  - daily updates for activation, differentiation, proliferation, and death.
- The logic matches the biological story we discussed:
  - activation from `N` during the antigen window,
  - **memory path** (S/C) vs **TEMRA path** (R),
  - **slow death for S/C**, faster for R.
- You convert the trajectories into a `DataFrame`, compute **total memory** and normalize to **percentages**, which makes the comparison to the EDA medians clear.
- The plots comparing:
  - `S` vs TSCM medians,
  - `C` vs TCM medians,
  - `R` vs TEMRA medians,
  - `S+C+R` vs total CD4 medians
    are exactly what I wanted.
- The `display_key_metrics_for_task4` helper is nice: it shows key days and makes it easier to check the qualitative behaviour.

Small suggestions

1. In the plots, your y-axis currently says "Cell Count" while you are plotting percentages. Please change the y-labels to something like **"% of CD4 cells (model)"** to avoid confusion.
2. Under the figure, add **a short text cell** (4–5 lines) explaining:
   - Does the model show low values at day 1, higher at day 22–43, and persistence at day 365?
   - Is TEMRA (R) smaller and more transient than TSCM/TCM?
3. It would be nice to show **at least one parameter sensitivity**:
   - for example, run the model with `frac_memory = 0.5` vs `0.8` and briefly describe how this changes the shape of S/C/R.

Overall: this **compartment model is well done** and is a good first mechanistic description of the CD4 data.

2. About "ABM": what you have and what is still missing

Right now:

- `Simple_ABM_CD4_TYF.ipynb` contains a **deterministic compartment model** (mean-field version).
- `ABM.ipynb` only has the Mesa imports, but no actual **Model/Agent** implementation yet.

So from a modelling point of view:

- You have done step 1: **compartment / ODE-like model**.
- The next step (for your Bachelor project) is step 2: a **true agent-based model** where each cell is an individual agent.

3. Next step: build a real agent-based model (per-cell)

For the ABM, I don't need anything complicated. Start with a **simple custom implementation** :

3.1 Define a CD4Cell class:

```
import random

STATES = ["N", "S", "C", "R"]  # Naive, TSCM, TCM, TEMRA

class CD4Cell:
    def __init__(self):
        self.state = "N"
        self.alive = True
```

Use a similar set of parameters as in your compartment model:

- `T_antigen`, `p_act`, `frac_memory`, `p_SC`, `p_CR`,
- `d_N`, `d_S`, `d_C`, `d_R`.

3.2 Fate decision (key mechanism)

When a naive cell is activated during the antigen window, it flips a **fate coin**:

- with probability `frac_memory` → it follows the **memory path** and becomes TSCM/TCM (`S`/`C`)
- with probability `1 - frac_memory` → it follows the **TEMRA/terminal path** and becomes `R`

In code this looks like:

```
if cell.state == "N" and antigen(t) and random.random() <
p_act:
```

```
# activation + fate decision
if random.random() < frac_memory:
    # memory path
    if random.random() < 0.6:
        cell.state = "S"  # TSCM
    else:
        cell.state = "C"  # TCM
else:
    # terminal / TEMRA-like path
    cell.state = "R"
```

This `frac_memory` parameter is the main knob that controls whether a subject is **memory-dominant** or more **TEMRA-skewed**.

3.3 Simulation:

- Start with e.g. **1000 cells**, all naive.
- For each day `t = 0 ... 365`:
  - For each **alive** cell:
    - If `state == "N"` and antigen is present and `random.random() < p_act`:
      - decide **memory vs TEMRA path** with probability `frac_memory`:
        - memory → become S or C (e.g. 60% S, 40% C),
        - else → become R.
    - If `state == "S"` and `random.random() < p_SC`: change to `C`.
    - If `state == "C"` and `random.random() < p_CR`: change to `R`.
    - Apply **death** with the appropriate `d_*` for that state.
  - At the end of the day, **count how many alive cells** are in N, S, C, R and store it.

9. Plot the **fractions of S, C, R over time** and check if the qualitative behaviour is similar to your compartment model and to the EDA medians.

4. For our next meeting

For the next time we meet, please try to have:

1. Your current compartment model cleaned up a bit:
   - correct axis labels,
   - 1 short interpretation paragraph under the plots.
2. A **first working ABM notebook** (it can be in the same or a new file) that:
   - defines `CD4Cell`,
   - runs a simulation with ~1000 cells from day 0 to 365,
   - plots N, S, C, R over time (counts or fractions).

It doesn't have to match the data perfectly yet. The goal is just to have a **working agent-based simulation** that uses the same biological rules as your compartment model.

If any part of this is unclear, note down your questions and we'll go through them together.